

MICROMOLD · REQUIREMENTS

MicroMold requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/micromold		
Source	docs/03-requirements.md @ d0d37e2		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from MMD-CAL-001; R14 redefined to the press excluding molds and R13 to name the hood (MMD-DDR-001 D8, D9)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against first-order estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

MicroMold requirements

These are the requirements for the concept, checked by calculation at TRL 3 in MMD-CAL-001. Targets are proposals for review, not yet validated with users, and will be revised after co-design sessions (see MMD-PRB-001). "Met on paper" means met by calculation, and "met by design" means met by a stated feature; neither is demonstrated on hardware. Two targets changed in v0.3 under MMD-DDR-001, whose items are adopted for TRL 3 pending Amish's review: R14 now covers the press without molds (D8), and R13 names the hood (D9).

The **reference part** used throughout is a 30 g HDPE test plaque, 64 x 50 x 6 mm, molded from washed, dried flake of 3 to 8 mm.

Table 1. Requirements.

ID	Requirement	Target	Status (MMD-CAL-001)	Verification (TRL 3 or later)
R1	Process the safe recycled thermoplastics	HDPE, PP, LDPE and PS flake of 3 to 8 mm; PVC excluded by procedure and labeling	Met by design; injection-grade flake needed (CAL E6)	Design review; material trials later
R2	Shot size	30 g or more of HDPE per injection	At risk: 34.2 g ideal, about 25 g if the fresh charge is not tamped	Swept-volume calculation; later weighed shots
R3	Injection pressure	8 MPa (80 bar) or more at the melt with 250 N or less on the handle	Met on paper: 8.9 MPa at 0.60 drive efficiency (12.4 MPa at 0.84); fill needs about 4.8 MPa	Force and friction calculation; later load cell reading
R4	Temperature control	Barrel and nozzle zones each settable from 150 to 260 °C, held within ± 5 °C at the thermocouple; independent cut-out above 280 °C	Met by design, unverified	Controller and heater datasheets; later logging
R5	Warm-up	Cold start to 220 °C in 15 min or less	At risk: 14.8 min	Heat capacity calculation; later timed start
R6	Throughput	8 or more reference parts per hour with one mold	At risk: 9.4 per hour, but the mold settles at about 92 °C in still air	Cycle-time estimate; later timed trials
R7	Mold envelope	Molds up to 150 x 120 mm footprint and 40 to 120 mm stack height; 40 cm ² or more projected area at full pressure	Met on paper: 45 cm ² ; stacks of 30 to 120 mm fit	Clamp force calculation; model
R8	Low-cost tooling	A two-plate aluminum mold for the reference part machinable on a manual mill or small CNC for \$100 or less	Met on paper: \$90 (indicative)	Quotation from a local shop
R9	Bench size and mass	Press footprint 350 x 300 mm or less, top of handle 1.1 m or less above the bench, mass 35 kg or less without the control box	Not met on mass: 38.5 kg; 320 x 260 mm and 1,089 mm met	Model and mass estimate
R10	Power supply	Single-phase 230 V or 120 V, 1 kW or less, 10 A or less at 120 V	Met on paper: 635 W, 5.3 A at 120 V	Heater and fan ratings
R11	Touch-safe outer surfaces	Guard and jacket 60 °C or less at 220 °C set point and 25 °C ambient; nozzle and mold zone guarded	Met on paper: skin 48 °C; nozzle zone shield added	Thermal calculation
R12	Electrical safety	Earthed frame, fused inlet, double-pole switch, RCD or GFCI supply, heaters and wiring rated for 250 °C at the barrel	Met by design, unverified	Design review against IEC 60204-1 principles
R13	Fume control		Met on paper: about 124 m ³ /h	

ID	Requirement	Target	Status (MMD-CAL-001)	Verification (TRL 3 or later)
		A side hood with a duct fan at the funnel, and operation only under it or outdoors; set points above 260 °C blocked in the controller		Hood calculation; design review
R14	Affordable	Parts for the press, excluding molds (tooling, see R8), \$400 or less	Not met: \$495 (\$585 with one mold)	Priced BOM (bom/ bom.csv)
R15	Garage-buildable	Everything except the barrel and the mold built with hand tools, a drill press and optional welding	Met by design: the barrel with its nozzle and the molds come from a local machine shop; the plunger is only cross-drilled	Build sequence review
R16	Repeatable parts	Part mass within ± 3 % over 10 consecutive shots	Not verifiable at TRL 3	Later bench trials (TRL 4, not in the current phase)

■ Assumptions

- HDPE melt density about 0.75 g/cm³ at 200 to 220 °C; solid density about 0.95 g/cm³ (typical handbook values, estimates). The other property values are in MMD-CAL-001, Table 1.
- Arbor press drive: pinion pitch radius about 20 mm, 450 mm ratchet handle, design efficiency 0.60 (0.84 calculated), to be checked on the chosen press.
- Processing temperatures follow published melt ranges: HDPE 190 to 230 °C, PP 200 to 240 °C, LDPE 220 to 245 °C, PS 180 to 260 °C ([RJC Mold](#)).
- Four M10 class 8.8 bolts preloaded to about 20 kN each clamp the mold plates, with a factor of 2 against melt pressure opening the parting plane.

■ Requirements not met or at risk

- **R14 cost:** \$495 for the press against \$400, even with molds counted as tooling; \$585 with the first mold. Options are in docs/REVIEW.md , "Proposed, awaiting Amish".
- **R9 mass:** 38.5 kg against 35 kg, with an assumed 8 kg arbor press head. Options are in docs/REVIEW.md .
- **R2 shot size:** met only if each top-up of flake is tamped with the press while loading.
- **R5 warm-up:** 14.8 min against 15 min; 300 W bands would give 12.2 min (proposed).
- **R6 throughput:** met on paper at 9.4 per hour, but in still air the mold runs at about 92 °C and parts may warp; a cooling fan is proposed.
- **R16** cannot be assessed until hardware exists (TRL 4, outside the current phase).